PS 4.2

Picking Floor Digital Twin

A 3D twin of a warehouse picking floor, driven by **2,293 real customer orders**. Re-shelve the products; it replays the same month and **measures** the walking saved.

2,293	−15.8%	−68%
real orders, 16,225 pick lines	walking, optimized layout	plus smart batching



Team GitHappens · SENSE, VIT
Rohith Krishna A K (25BCE2027) · Jaidev A (25BCE0971)
Guide: Dr. Jaffino

§01 THE PROBLEM

Pickers walk more than they pick.

53%

79 s walking | 71 s picking

per order · 1.2 m/s · 10 s per line

218 km a month for one picker in a 100-slot zone.

§02 REAL DATA

Online Retail II (UCI, CC BY 4.0): UK gift wholesaler, November 2010, pre-Christmas peak.

Raw order lines	76,584
Clean pick lines	74,757
Top 100 products	16,225
Orders replayed	2,293

Top 100 = 3% of the range, 22% of picks. 7.1 lines/order.

§03 HOW IT WORKS

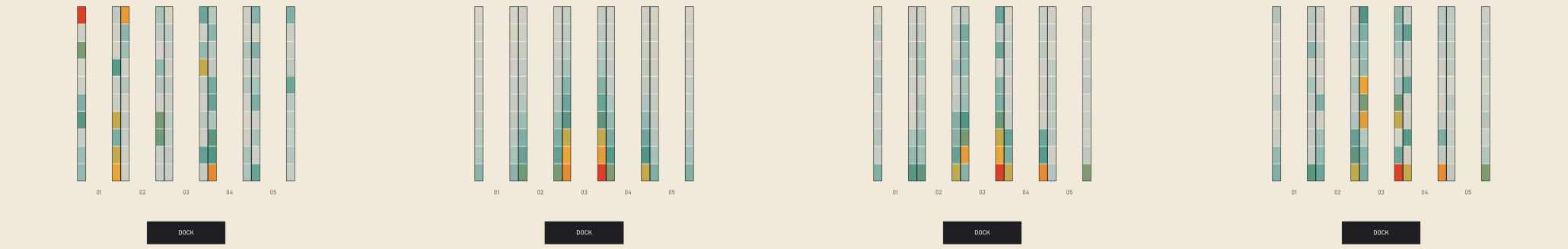
01 **Clean** — pandas removes cancellations, fees and bad rows

02 **Floor graph** — 61 nodes, 64 edges; no edge crosses a rack

03 **Route & score** — nearest neighbour + 2-opt; a month in ms

04 **Optimize & play** — same code in Node and the browser

§04 THE LAYOUT LADDER: SAME 2,293 ORDERS, FOUR WAYS TO SHELVE



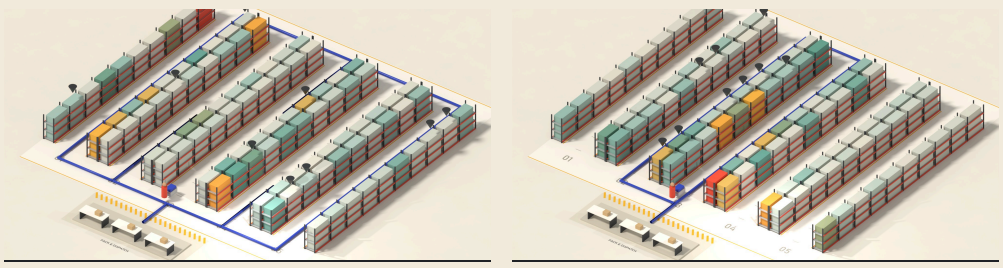
Original 218.1 km
AS RECEIVED
Stock-code order: the baseline.

ABC −2.5%
INTUITION
Most-picked nearest the dock. Orders still enter most aisles.

Affinity −11.3%
DATA INSIGHT
Products bought together share an aisle.

Optimized −15.8%
ALGORITHM
Simulated annealing: 80,000 swaps, each judged by exact replay.

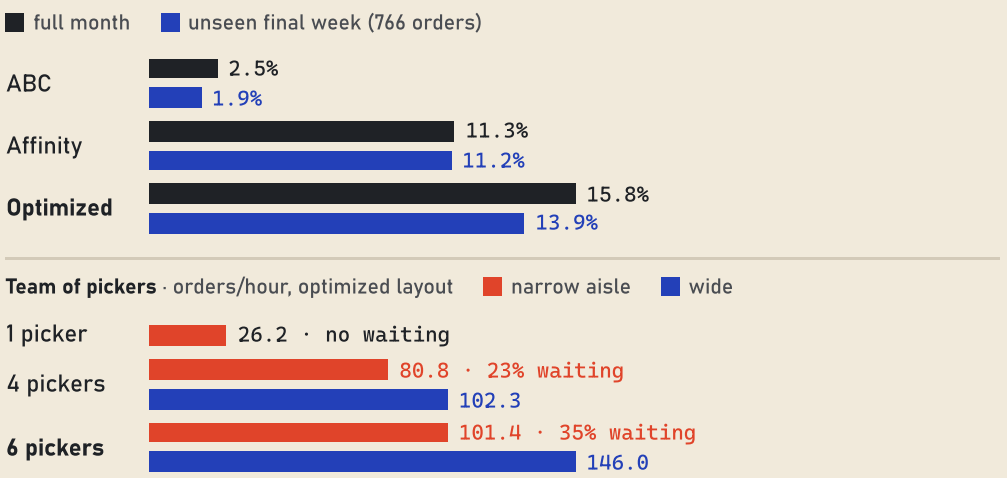
§05 SAME ORDER, TWO FLOORS



ORIGINAL · 5 AISLES 143 m | OPTIMIZED · 2 AISLES 62 m

Invoice 532964, 9 lines: one of the month's biggest wins. On average 95.1 → 80.1 m per order.

§06 PROOF & A TEAM OF PICKERS



§07 PLAY IT: CAN YOU BEAT THE OPTIMIZER?



§08 KEY FINDING

The textbook ABC fix saves only **2.5%**: orders average 7 lines, so pickers still visit most aisles. Using **what is bought together** saves 4–5× more, and it holds on orders the model never saw.

01 **Swap racks** · the whole month re-scores instantly

02 **Live optimizer** · annealing in a Web Worker; 182.3 km in 14 s

03 **Team mode** · 1–6 pickers queue at full aisles; slotting gain shrinks 9% → 5%

04 **IoT sensor mode** · RFID gate, UWB cart tag, pick-to-light, barcode scans, as JSON Lines

PYTHON · PANDAS · NODE · THREE.JS · VITE · BLENDER (CYCLES BAKE + RENDERS) · RUNS FULLY OFFLINE

DATA: ONLINE RETAIL II, UCI, CC BY 4.0